Morris Darren Babu

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CAREER HIGHLIGHTS

Structured AI/Data Specialist skilled at transforming and automating data workflows. Proficient in AWS, Azure, and Docker for secure cloud-native DevOps. Expert in analytics, ML/DL, and interactive visualization for data-driven decisions. Experienced in GenAI, AI Agents and LLMOps, including LLM evaluation, prompt engineering, RAG, and LoRA/QLoRA fine-tuning. Drives innovation through agile collaboration and Industry 4.0 frameworks.

PROFESSIONAL EXPERIENCE

10/2025 – 03/2026
Bonn, Germany

Deutsche Telekom, *MLOps Engineer Intern | LLM Infrastructure & Deployment*

- Built a microservices-based LLM benchmarking framework on vLLM using GuideLLM (latency, throughput) and DeepEval (LLM-as-judge), tracking experiment runs in ClearML to define SLOs for model selection and capacity planning.
- Scaled automated benchmarking via GitLab CI/CD and Kubernetes runners, executing containerized matrix jobs on Hugging Face datasets, publishing reports to Grafana.
- Secured enterprise MLflow deployments on Kubernetes, implementing Keycloak OIDC and OAuth sidecars for SSO, enforcing ingress, TLS, and RBAC to meet compliance.

05/2025 – 09/2025
Wolfsburg, Germany

Volkswagen Group (CARIAD), *Master Thesis | Integrating LLMs In Automotive Systems*

- Implemented AI-driven black-box fuzzing in Azure AI Foundry by benchmarking 16 LLM models, boosting code-flaw detection by 13%, strengthening ECU resilience, and reducing OEM cybersecurity threats.
- Automated test-case generation by containerizing LLM inference services in CI/CD/CT pipelines, cutting test creation time by 33% and increasing code coverage by 7%.
- Designed a secure Azure Private Link for an automated LLM fuzz-testing PoC, replacing manual tests, reducing delays by 10%, and improving automotive software quality.

10/2024 – 04/2025
Munich, Germany

BMW Group, *Data Scientist Intern | RAG & LLM Systems*

- Deployed a CRNN surface detection model on AWS SageMaker (EC2, S3) with semantic segmentation, tuning AI parameters and serving results via API to cut false positives.
- Architected a Django Web application for parsing geometric STL meshes and file ingestion, utilizing document chunking pipelines to achieve 8% faster processing.
- Engineered an LLM auto suggestion model utilizing RAG and vector search to identify part issues, via document chunking and data indexing to elevate precision by 12%.

10/2021 – 09/2022
Chennai, India

Crystal Consultancy Services, *Data Analyst*

- Monitored and analyzed key performance indicators (KPIs) using SAP R3 data, resulting in a 4% improvement through targeted data-driven decisions.
- Applied Azure Databricks to address slow data processing in the ETL pipeline, optimizing feature engineering and model deployment, enhancing speed by 25%.

02/2021 – 07/2021
Texas, USA
(Remote)

Altascio Technologies, *Data Engineering Intern*

- Utilized Excel techniques (Pivot, VBA, VLOOKUP, SUMIFS) to streamline manual aggregation of department spreadsheets, reducing report generation time by 2%.
- Streamlined data extraction and consolidation using SQL to resolve fragmented inventory data, saving 20 hrs/week and reducing inventory update time by 11%.

06/2020 – 12/2020
London, United Kingdom (Remote)

Jman Group, *Python Developer Intern*

- Assisted in developing predictive models with ML algorithms using sensor data to identify clusters, detect outliers and trends, delivering 87% accuracy in field tests.
- Applied end-to-end data science (data collection, EDA, feature engineering, model selection, deployment) to solve ETL pipeline bottlenecks, raising throughput by 27%.

TECHNICAL SKILLS

Python (Scikit-learn) | R | C++ | JavaScript | Computer Vision | TensorFlow, PyTorch

GenAI (Prompting, RAG, GAN, VAEs, LangChain) | SQL | Snowflake | Power BI | MS Office

Azure (Spark, Data Factory, Data Lake) | AWS (EC2, Aurora, CloudWatch) | Keycloak | n8n

Git | GitHub Actions(CI/CD) | Docker | Kubernetes (Helm) | MLflow | SAP S/4 HANA

SOFT SKILLS

Analytical / Strategic Thinking | Creative | Communication | Problem Solving | Teamwork

EDUCATION

10/2022 – 03/2026

M.Sc. in Data Science, *Friedrich Alexander University Erlangen-Nuremberg*

Subjects : Pattern Recognition, Deep Learning, Generative Artificial Intelligence, Anomaly Detection, Decision Theory, Reinforcement Learning, Statistics, ML in Finance.

06/2018 – 06/2022

B.E. in Computer Science Engineering, *Rajalakshmi Institute of Technology, India*

PROJECTS

IntentGraph (Agentic AI Workflow Orchestration System)

Built a agentic platform that compiles natural-language goals into structured, previewable workflows with risk analysis, policy checks, human approvals, durable execution, and rollback across APIs and developer tools.

Material Prediction & Mesh Dataset Generation with GNNs (Graph Neural Networks)

Constructed a custom GNN framework and synthetic dataset of shape/material data, predicting wear with RMSE under 0.1 and supporting end-to-end shape generation and graph visualization.

OCR with Handwritten Recognition and Auto Spelling Correction, *Bachelor Thesis*

Implemented an NLP-driven spelling correction system using tokenization, language modeling, and Tesseract OCR, exposing it via a scalable REST API and reducing Character Error Rate by 89% on multilingual documents.

Movie Recommendation System using Hybrid Filtering, *Publication*

Created a hybrid recommendation system using collaborative and content filtering with clustering, achieving a 92% Top-K Hit Rate and integrating it into a Flask-based web app for real-time user recommendations.

CERTIFICATIONS

Predictive Data Analytics for Business Intelligence Nanodegree | Excel to MySQL: Regression | Statistics | DL AI TensorFlow Developer | Google Cloud Platform IT Automation | Data Warehousing, Big Data on GCP

LANGUAGES

English: C1 (Business fluent) | **German:** A2 erworben, B1 laufend | **French:** A1 | **Tamil:** Native